

Bucket Wheel Excavators

It is a multi-bucket continuous excavator: continuous digging with this m/c is done with evenly spaced buckets fitted on the periphery of a rotating wheel (cutting wheel). Once the material is dug by the buckets, it flows in a continuous stream via transfer to the conveyor system that in turn transports the material to the discharge point. The wheel is adjusted in rpm mounts 6 to 20 buckets of equal size depending on the size and model of the m/c. BW's may be used for both cutting - above its crawler level (high cut) and below its crawler level (deep cut) depending on the m/c design & though BW's are more frequently used for high cut.

The BW were first developed to handle only the easy digging materials such as brown coal (lignite) in Germany.

But with the advancements in electric drives, metallurgy of bucket & teeth and introduction of high steel cord belt the BWEs are finding their application in digging medium to medium-hard digging materials, such as hard pottery clays, oilshales etc.

BWEs are among the largest terrestrial vehicles ever constructed - the biggest one built the Tenova - TAKRAF SR 8000 has a weight of 14,200 ton and moves 240,000 m³ of OB per day. Tenova TAKRAF BWE have a wide range of capacities (200 to 16000 m³/h) and can excavate bench heights from less than 5m to max. 51m.
(www.takraf.com)

Classification of BWE:

1. Based on Bucket type, the BWE may be classified into 3 categories.
 - a) Cell-less type - most popular.
 - b) Semi Cell type.
 - c) Cell type.
2. Based on propel mech:
 - a) Crawler ~~for~~ mounted. (3-point crawler support)
 - b) Rail mounted.
3. Based on type of Cattle Boom: (The boom can be extended).
 - Non-crowding.
 - crowding.

4. Type of discharge boom

Non telescopic type

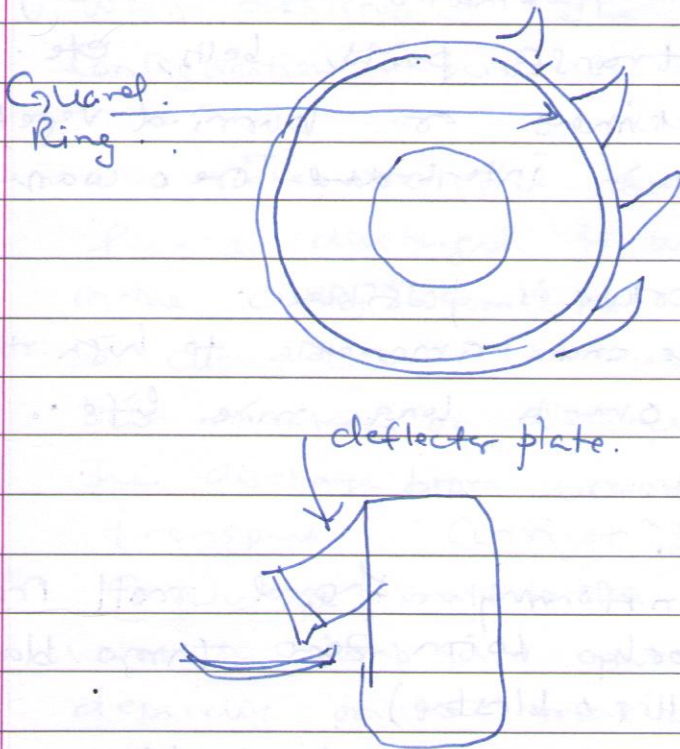
Telescopic type

Direct castor type

5. Cutting/digging

- High digging type

- High & deep digging type



6. Individual Bucket cap & digging force

Individual Bucket cap = 350 kN, 700 kN

The undercarriage of the mc is supported by 3-sets. (3×4 or 3×6) of wide crawlers. The undercarriage supports the revolving frame that in turn supports the Super Str.

Applicability & Limitation:

Applicability

- Formations are Soft to med hard
- free of geologic disturbance
- free of boulders
- Even though the Formations will not clog the cutting wheel, transfer points, belts, etc. because of stickiness or buried vegetation
- selective mining of interbedded ore & waste is necessary.
- continuous working is possible.
- reserve is large and amenable to high rate of extraction over a long mine life.

Limitation -

- presence of uniform hard mat that fragments poorly with zero throw blasting (limited drilling & blasting).
- high % of large & un-sized boulders that is too big for the buckets to handle during cutting.
- presence of buried vegetation or such things that would clog the cutting wheel, belts, transfer points, etc.
- presence of interbedded dirt & sand.

USAID pdf. → Presentation. on. NLC By Chief Mgo (Mining).

Operation of BWE:-

A BWE is a continuous type excavator designed for cutting of soft to med. hard digging strata that are amenable to excavation without blasting or with the help of zero-thrust blasting in some cases. The BWE's are generally used for large capacity excavation over long mine life from higher benches. The m/c. is generally used to dig above its crawler level (high cut) but can also dig below its crawler level (deep cut) when designed so. The mining face and bench configuration depends on the m/c. designed parameter.

The material dug by the buckets from the face is discharged to the belt conveyor mounted on the cutting boom of the m/c. The matl carried by the cutting boom conveyor is transferred to the belt mounted on discharge boom via a transfer chute. The discharge boom transfers the matl to the mine transport conveyor system. The conv. system.

Carries the matl to the dump yard or stockyard — waste to dump yard and ore to stockyard — depending on the material being excavated. The bench conveyor needs to be shifted periodically depending on the advancement of the face. Sometimes a mobile transfer conveyor is used to reduce the frequency of bench conv. shifting.

Dragline.

1. Basic m/c.
2. & additional components.
3. Erection 3 or 4 years.

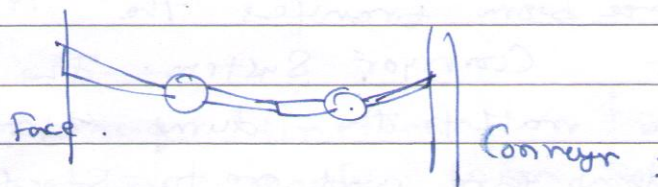
BWE

1. Designs are ready.
2. Float a Global tender.
3. Specifications discussed with Suppliers.
4. Tenders opened in front of representatives of the co.
5. Negotiations. (Open negotiations).
6. Study of field condn. (3-6 months).
7. Design modified.
8. Parts mfgd. (3 to 4 years).
9. M/C testing on short term basis.
10. Part payment made on basis of productivity.

BWEs are mfgd ^{on} demand and order.

For 50 to 100 m. Shrinkage (1 to 2 shifts) Conveyor Shifting.

Mobile Transfer Conveyor $\rightarrow$ Boom Swing $\pm 90^\circ$



The Conveyor System that carries the material to the dump yard may either be along the pit system (APS) or cross pit system (XPS). When XPS is adopted conveyor bridges that span across the pit to the backfill required to be provided.

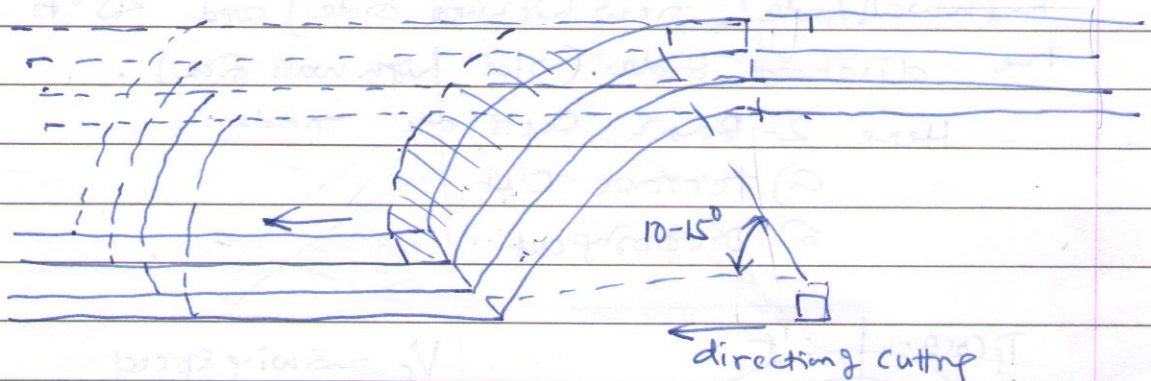
For dumping the matl carried by conveyers either Spreads or stackers are used. In the

Method of operation:-

- 1.) Lateral or half block method.
- 2.) Full block method of excavation.

Lateral / Half Block Method:-

The m/c. travels parallel to the highwalls with Crawler offset. The m/c. cuts long deep benches or terraces along the highwall keeping both the highwalls (old & new) on the same side, and hence the method is also known as $\frac{1}{2}$ block method:



Height terrace = 0.5 to $0.75 D$. (D = dia of Bucket).

The material is cut slice by slice.

In this method the excavation the angle of Swing of the cutting boom with the highwall is from 10° to 15° through 90° to the direction of travel. For this operation the BWC generally requires to be equipped with long cutting boom.

The m/c. cuts the whole bench at in regd no of slices starting at the top. To start with the m/c. engages the forward or cutting wheel.

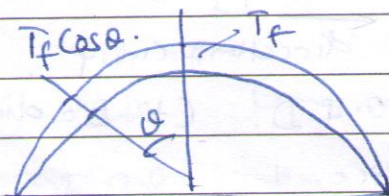
There is significantly more travel time is reqd that with the full block method; reducing the actual digging time and increasing the crawler wear. However, the method is mostly used in case of selective mining of inter-bedded ore and waste.

Full-Block Method:-

The BWC travels $\parallel$ to high wall removing a block width that is $\approx$ of wheel cutting radius and swing arc. Optimum block width is normally achieved with a swing upto 80° to the high wall side (new high wall side) and 45° to 55° on the discharge side (old high wall side).

Here 2-Basic cuts are made.

- terrace cut
- Dropping cut.



V_s = Swing speed

$$\text{Vol of Cuttip} = V_s \times T_f$$

$$V_s T_f \cos \theta$$

The Swing ~~cut~~ speed should increase by $\frac{V_s}{\cos \theta}$

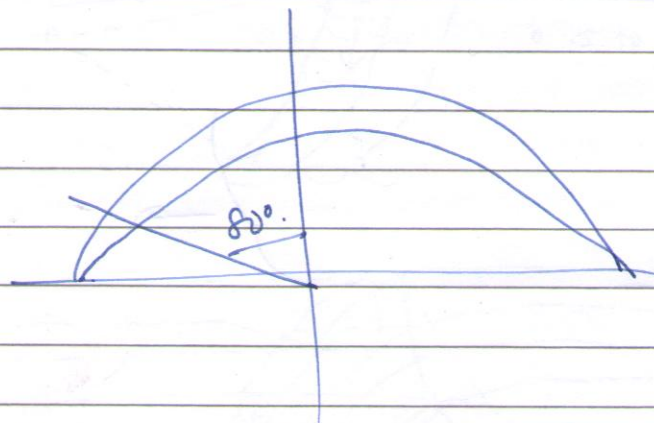
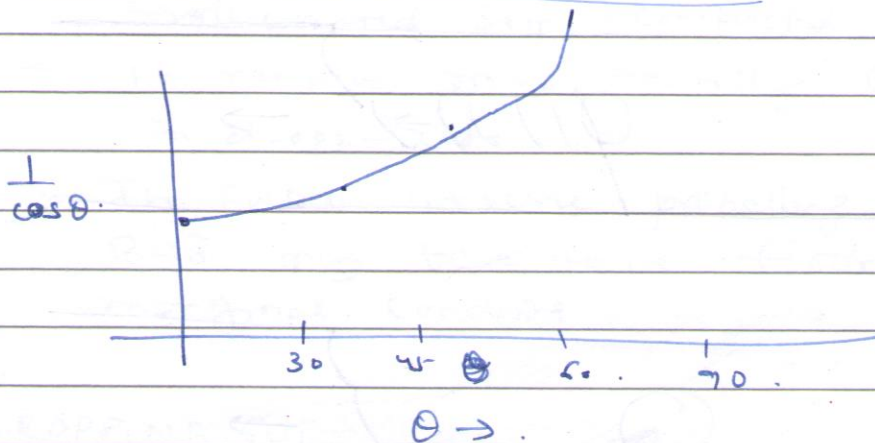
so that all through

$$\text{it is } \frac{V_s T_f \cos \theta}{\cos \theta} = V_s T_f$$

$$\forall \theta = 30^\circ \quad \cos \theta = \frac{\sqrt{3}}{2} = \frac{1.732}{2} = 0.85$$

$$\frac{1}{\cos \theta} = \frac{2}{\sqrt{3}} = 1.17$$

θ	$\cos \theta$	$\frac{1}{\cos \theta}$
30°	0.85	1.17
45°	0.7	1.41
60°	0.5	2
90°	0	∞

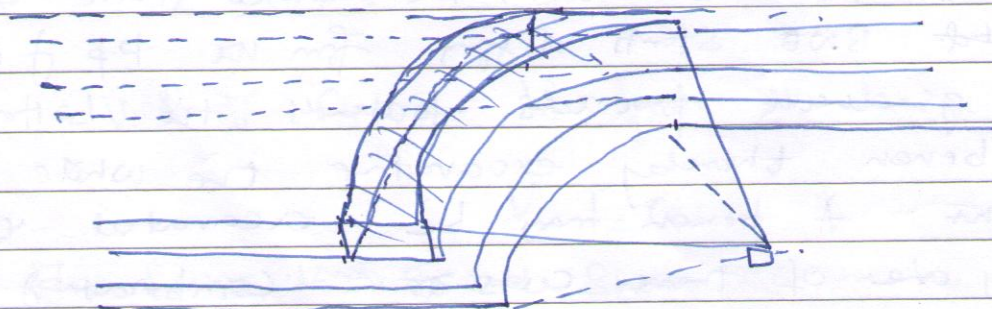
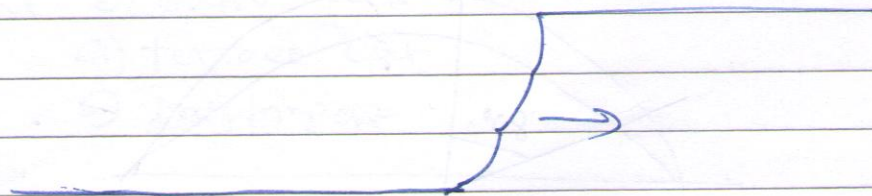
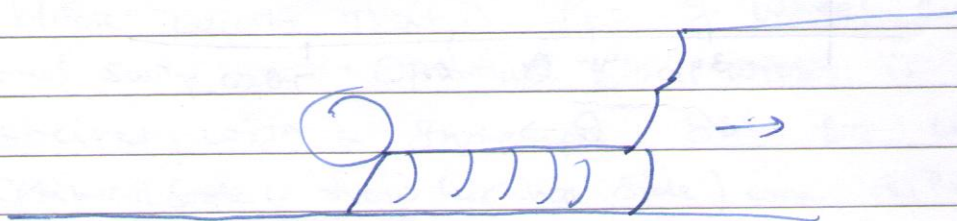
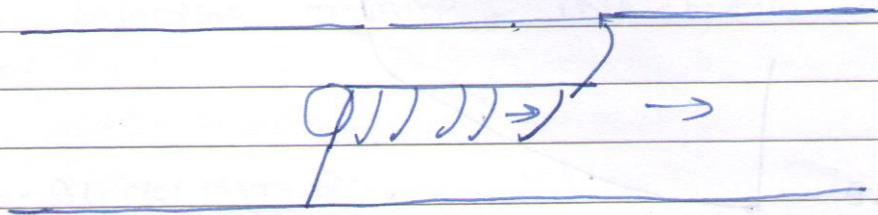
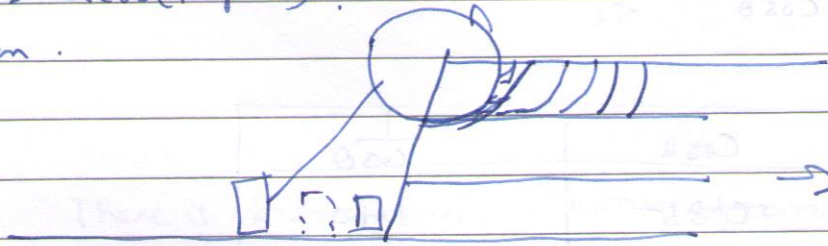


In both the cuts, the ~~stage~~ face slope is established. BWS starts digging from the top of bench and gradually proceeds towards the bottom of the bench thereby excavating the whole bench height. A bench may be excavated either by any one of these cuts or a combination of these cuts. FULL BLOCK IS USED WHEN SELECTIVE MINING IS NOT REQUIRED.

Terrace Cut:-

Cut & Slew (in plan)

↓
section

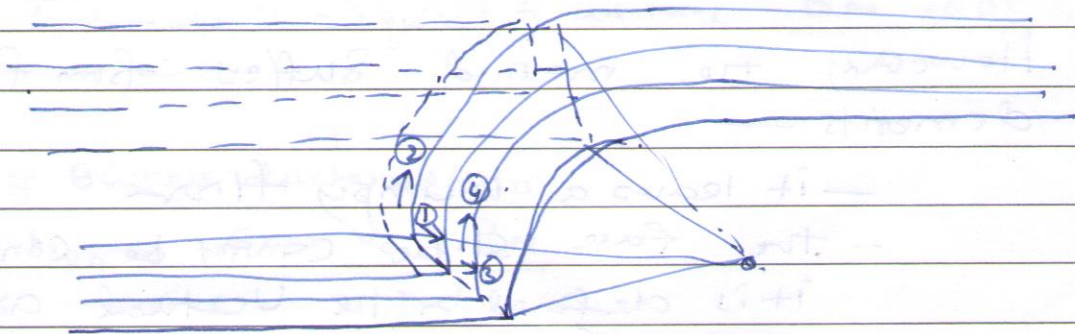


Merits:

- it leaves a comparatively level floor.
- face slope can be controlled by maintaining design slope between two successive terraces.
- For the same BWE it can excavate higher bench compared to by dropping cut.

Demerits:

- tends to produce large lumps when digger into agglomerated semi-compacted soil.
- it requires more travelling compared to that in dropping cut.
- The FWD reverse propelling sequence of the BWE may tear up the floor under marginal conditions.

DROPPING CUT:-

The m/c is not coming and going back in this method.

The final slope of face defined by the vertical arc of the cutting wheel about its boom pivot point on the m/c. main frame. cuts are made by the bottom arc of the cutting wheel progressing from the top to the bottom of the bench face as the wheel is slewed from side to side. To start the cut separate the bottom arc of the cutting wheel is placed at one end of the face to create the first cut height and then the wheel is slewed from side to side.

Once a cut is made over the whole face width, the cutting wheel is dropped down for another.

Merits:-

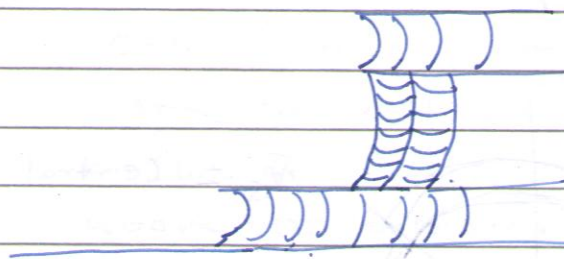
- minimizes lumps and improves η in haul out of a stable high wall.
- provides easier operator control of the cutting slice depth and overall slope.
- Requires less m/c. travel compared to that in terrace cut.

However, the method suffers from following demerits.

- it leaves a lumpy floor
- the face slope cannot be controlled, it is defined by the vertical arc of the cutting wheel.
- it normally requires more power compared to that in terrace cut.
- For the same BWG it cannot excavate as high benches as by terrace cut.

Combination of Terrace cut & Dropping Cut:-

to avoid bumpy floors.



$$\omega = \frac{V}{r} = \frac{V}{R}$$

$$V = r\omega$$

Top by terrace, bottom by terrace and in between by drop cut.

Calculation of Productivity of BWS:-

Consideration:-

— How many buckets are discharged per unit time?

$$Q_{th} = I \times N_{BD} \times S_f \times 60 \frac{m^3}{hr}$$

I = individual bucket capacity. (Normally lt , convert to m^3)

N_{BD} = No of buckets discharging per unit time

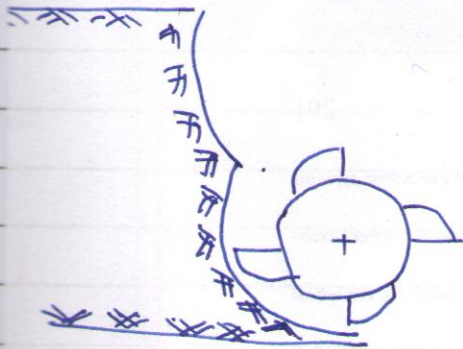
$$S_f = \text{swell factor} = \frac{100}{100 + S} \quad \text{if } S = \% \text{ swell.}$$

$$N_{BD} = \frac{60 \times V_c \times B_N}{\pi D}$$

V_c = wheel speed (tangential speed at cutting edge)

B_N = No of buckets attached to wheel (6 to 20)

D = dia of cutting wheel in (m)



Solution for the Assignment Problem.

Given.

B_N = No of Buckets in the BWE.

v_c = velocity of tip of Bucket (m/s).

D = Diameter of Cutting wheel (m).

The problem is to find "The number of buckets discharged per unit time" = ?

Time taken to discharge B_N buckets.

= time taken for full rotation.

$$= \frac{2\pi}{\omega} = \frac{2\pi \times (\frac{D}{2})}{v_c} \quad \text{But } \omega = \frac{v_c}{\text{Radius}}.$$

$$= \frac{\pi D}{v_c} \text{ Sec.}$$

$$\therefore \text{time taken to discharge } B_N \text{ Buckets} = \frac{\pi D}{v_c} \text{ sec}$$

$$= \frac{\pi D}{60 \times v_c} \text{ minutes.}$$

Now:

$$\frac{\pi D}{60 \times v_c} \text{ minutes are required to discharge } B_N \text{ buckets.}$$

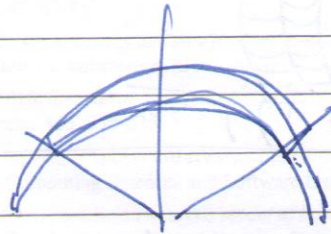
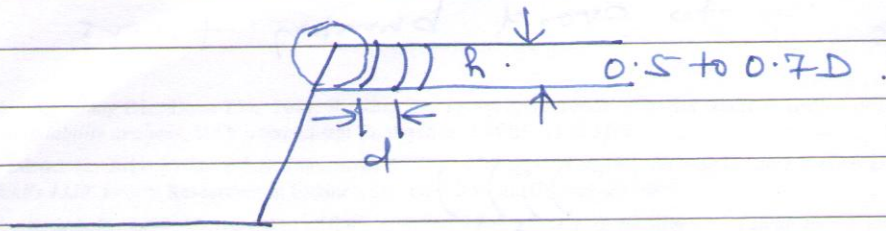
$$\Rightarrow 1 \text{ minute is required to discharge } \frac{B_N \times 60 \times v_c}{\pi D} \text{ buckets.}$$

$\therefore$ No of Buckets discharged/min

$$N_{BD} = \frac{60 \times v_c \times B_N}{\pi D}.$$

D = dia of Cutting wheel in (m).

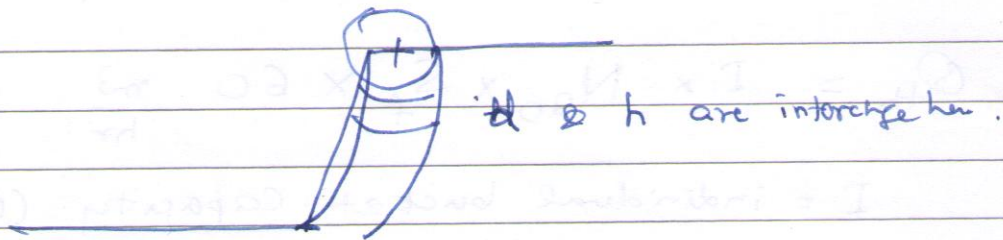
Imagine a terrace cut:-



$V_s = \text{Central speed (m/s)}$

$d = \text{cut depth. (Slice depth)}$

$$Q_1 = V_s \times D \times 3600 \times h. \quad \left(\frac{\text{m}^3}{\text{hr}} \right)$$



d & h are interchangeable.

$$Q_2 = V_s \times d \times h \times 3600. \quad \left(\frac{\text{m}^3}{\text{hr}} \right)$$

Use Q_1 or Q_2 whichever ever is lower.

V_s cannot be increased much. (Centrifugal force).

Advantages:-

- more Stable post Slopes for both highwall.
- o Spoil heap.
- availability of wide working spaces.
- Greater amt of developed reserves exposed by operating on wide benches.

- Spoil piles are more even in size and contour than with other excavators.
- Backfilled Spoil piles are more even in size and contour than with other excavators.
- Close control in selective mining of interbedded ore and waste (in lateral block method)
- ability to deliver material on the same, above or below the working level.
- high & deep cuts can be handled depending on the m/c design.
- comparatively smooth & level working benches.
- cost of excavation of matl/ton or/m³ is generally lower.
- min load fluctuating due to uniform operation.
- lone fleet of mobile transport eqpt requiring a commensurate manpower can be totally avoided.

Disadv

- High capital investment
- Requires high reserve, should be amenable to high rate of extraction over a long mine life.
- System of excavation is rigid in the sense that it ~~is~~ is not amenable to flexibility of operation compared to other conventional mining with shovel & dumper.

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